Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1	1 – 1

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0031813361611877	0.9947478991596639 – 1.4287222808870117

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0021276595744681	0.9978768577494692 – 1.004255319148936

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0005473453749316	0.9873041599135602 – 1.155377946356001

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0005652911249294	0.9435931601154786 – 1.1982515510434293

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.9245283018867925	0.9074074074074074 – 0.9433962264150944

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0043859649122806	0.6431924882629108 – 1.1081277213352685

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0027510316368637	0.9773936170212766 – 1.0097087378640777

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.7142857142857143	0.6944444444444444 – 0.7222222222222222

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.6666666666666666	0.6666666666666666 – 0.75

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0004301075268818	0.9575835475578406 – 1.0379633765073693

Cholesky factorization

256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9966810966810967	0.9867199771526489 – 1.0048934945308001

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0021162437741817	0.9961633557848069 – 1.0049226291120006

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9976331360946745	0.9316427783902976 – 1.005966587112172

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0054669703872436	0.9654383154039826 – 1.0339377033937704

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.99673735725938	0.9522240527182867 – 1.0662020905923344